Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

Here âĢĻs a concise solution outline:

- Kinetic energy: E = 1 eV âīĪ 1.602 ÃĹ 10^-12 erg.
- Relate kinetic energy to momentum using E = p^2/(2m), solving for p gives p = âĪļ(2mE).
- Mass of electron: m_e âīĪ 9.109 ÃĹ 10^-28 g.
- Compute momentum: p = âĪļ[2 ÃĹ (9.109 ÃĹ 10^-28 g) ÃĹ (1.602 ÃĹ 10^-12 erg)] âīĪ 4.36 ÃĹ 10^-21 gÂ·cm/s.
- De Broglie wavelength: Î» = h/p.
- PlanckâĢĻs constant in the appropriate units for cm: h âīĪ 6.625 ÃĹ 10^-27 ergÂ·sec.
- Thus Î» âīĪ (6.625 ÃĹ 10^-27) / (4.36 ÃĹ 10^-21) âīĪ 1.52 ÃĹ 10^-6 cm âīĪ 1.23 ÃĹ 10^-7 cm.

Therefore